

CO2 Assessment Tool 1 – Scenario-Based Requirement Analysis

Software Requirement Specification (SRS)

AI InterviewX – AI-Powered Mock Interview Platform

Field	Details
Course Code	CSA1012 – Software Engineering
CO Assessed	CO2: Perform detailed requirements analysis, design scalable architectures, and prioritize features
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Project Title	AI InterviewX – AI-Powered Mock Interview Platform

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for AI InterviewX, an AI-powered mock interview platform designed to help job seekers practice technical, HR, and coding interviews and receive automated, data-driven feedback. It is intended for use by the development team, project evaluators, and stakeholders to establish a shared understanding of system scope and behaviour.

1.2 Business Domain

EdTech / Career Services – specifically AI-assisted interview preparation and skill assessment for engineering and computer-science students and early-career job seekers.

1.3 Intended Audience

- Development team (frontend, backend, and AI/ML engineers)
- Project evaluators / course faculty (CSA10 – Software Engineering)
- Placement cell / recruiters using the analytics module
- End users – candidates preparing for interviews

2. Problem Description

Final-year students and job seekers often lack structured, repeatable opportunities to practice interviews before facing real recruiters. Peer mock interviews are inconsistent, human interviewers are not always available, and generic question banks do not adapt to the candidate's domain, skill level, or communication gaps. There is a need for a platform that can simulate realistic interview conditions, ask domain-relevant questions, evaluate spoken/written answers using AI, and provide immediate, actionable feedback – without depending on the availability of a human interviewer.

3. Scope of the System

AI InterviewX will be a web-based platform that allows candidates to register, choose an interview domain and role, undergo an AI-simulated mock interview (technical, HR, and coding rounds), and receive an automatically generated performance report with a score and improvement suggestions. Recruiters/admins can manage the question bank, configure evaluation rules, and view aggregated candidate analytics.

Out of scope for the current release: live human-proctored interviews, integration with external Applicant Tracking Systems (ATS), and payment/subscription billing — these are identified as possible future enhancements.

4. Stakeholders and System Objectives

- Candidates (job seekers) – primary users seeking interview practice and feedback.
- Admins / Recruiters – manage content and monitor candidate readiness.
- Institution / Placement Cell – uses aggregate analytics to gauge student readiness.
- AI/ML Engine – internal system component that generates and evaluates interview content.

System objectives: (1) simulate realistic, domain-specific interviews on demand, (2) evaluate responses objectively using NLP/ML, (3) give candidates actionable feedback to improve, and (4) give institutions visibility into candidate readiness at scale.

5. Functional Requirements

ID	Requirement	Priority	Related Actor
FR-01	The system shall allow candidates to register and log in using email/password or OAuth.	High	Candidate
FR-02	The system shall let candidates select an interview domain (e.g., Software Engineering, Data Science, AI/ML) and target role.	High	Candidate
FR-03	The system shall generate AI-driven interview questions (technical, HR, coding) based on the selected domain and difficulty level.	High	AI Engine
FR-04	The system shall capture candidate responses via text, audio, or webcam-based video input.	High	Candidate
FR-05	The system shall use an NLP/ML model to evaluate answers for correctness, relevance, and communication quality.	High	AI Engine
FR-06	The system shall score candidate performance and generate a detailed feedback report after each session.	High	AI Engine
FR-07	The system shall allow candidates to view historical interview attempts and track progress over time.	Medium	Candidate
FR-08	The system shall allow admins/recruiters to add, edit, or remove questions from the question bank.	Medium	Admin
FR-09	The system shall allow admins to configure AI evaluation rules and scoring weightage.	Medium	Admin
FR-10	The system shall allow admins to view aggregated analytics of candidate performance across domains.	Medium	Admin
FR-11	The system shall allow candidates to download their performance report as a PDF.	Low	Candidate
FR-12	The system shall send email/SMS notifications for scheduled mock interviews and results.	Low	Candidate

6. Non-Functional Requirements

Category	Requirement
Performance	The system shall evaluate a candidate's response and return AI feedback within 5 seconds under normal load.
Scalability	The system shall support at least 5,000 concurrent mock-interview sessions without degradation.
Security	All candidate data, video/audio recordings, and reports shall be encrypted in transit (TLS 1.2+) and at rest.
Reliability	The platform shall maintain 99.5% uptime and gracefully handle AI-model or network failures with retry logic.

Category	Requirement
Usability	The interface shall be intuitive enough for a first-time user to complete a mock interview without external help.
Maintainability	The AI evaluation module shall be loosely coupled from the core application so models can be retrained/updated independently.
Availability	The system shall be accessible 24/7, with scheduled maintenance windows communicated in advance.
Portability	The web application shall function consistently across major browsers (Chrome, Edge, Firefox, Safari) and be responsive on mobile devices.
Compliance	The system shall comply with applicable data-protection regulations (e.g., IT Act 2000 / GDPR principles) for storing candidate PII.

7. Actors and Use Cases

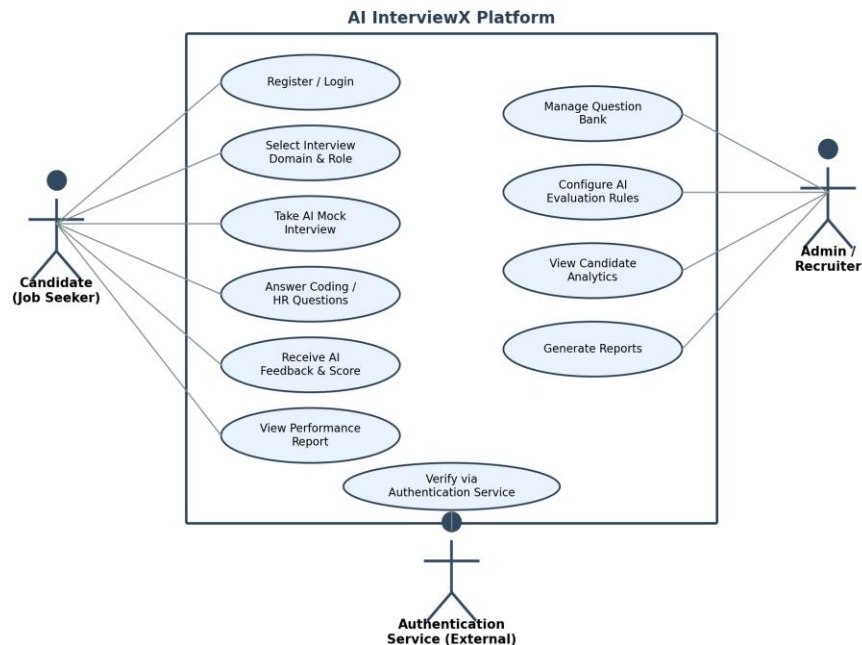
7.1 Actors

- Candidate (Job Seeker) – primary actor who takes mock interviews.
- Admin / Recruiter – primary actor who manages content and reviews analytics.
- AI Engine – internal secondary actor that generates questions and evaluates answers.
- Authentication Service (External) – secondary actor verifying user identity.

7.2 Use Case Table

Use Case	Actor	Description	Pre-condition
UC-01 Register/Login	Candidate	Candidate creates an account or logs in to access the platform.	Candidate has internet access
UC-02 Select Domain	Candidate	Candidate selects interview domain, role, and difficulty level.	Candidate is logged in
UC-03 Take Mock Interview	Candidate	Candidate answers AI-generated technical/HR/coding questions in a timed session.	Domain selected
UC-04 Receive AI Feedback	Candidate, AI Engine	AI engine evaluates responses and returns a score with strengths/improvement areas.	Interview session completed
UC-05 View Performance Report	Candidate	Candidate views/downloads a consolidated report of past interview attempts.	At least one interview completed
UC-06 Manage Question Bank	Admin	Admin adds, updates, or removes interview questions and model answers.	Admin is authenticated
UC-07 Configure Evaluation Rules	Admin	Admin sets scoring weightage and evaluation parameters for the AI engine.	Admin is authenticated
UC-08 View Candidate Analytics	Admin	Admin views aggregated performance analytics across candidates and domains.	Sufficient interview data exists
UC-09 Verify via Authentication Service	Candidate, Admin, Auth Service	External identity provider verifies user credentials during login/registration.	Auth service is reachable

7.3 Use Case Diagram



8. Data Requirements

The system requires the following core data entities to support interview generation, evaluation, and reporting:

Data Entity	Key Attributes
Candidate Profile	CandidateID, Name, Email, Phone, Target Role, Resume/Skills, Registration Date
Interview Session	SessionID, CandidateID, Domain, Difficulty, StartTime, EndTime, Status
Question Bank	QuestionID, Domain, Type (Technical/HR/Coding), DifficultyLevel, ModelAnswer, Tags
Response Record	ResponseID, SessionID, QuestionID, CandidateAnswer (text/audio/video ref), Timestamp
Evaluation Result	ResultID, ResponseID, AIScore, Feedback Text, Confidence, EvaluationParameters
Performance Report	ReportID, CandidateID, SessionIDs[], OverallScore, StrengthAreas, ImprovementAreas, GeneratedDate
Admin/Recruiter Profile	AdminID, Name, Email, Role, Permissions

9. External Interface Requirements

9.1 User Interfaces

- Responsive web application (candidate portal and admin dashboard) accessible via desktop and mobile browsers.

9.2 Hardware Interfaces

- Candidate's webcam and microphone for video/audio-based mock interview rounds.

9.3 Software Interfaces

- Third-party Authentication/OAuth provider (e.g., Google Sign-In) for login.
- Speech-to-text API for converting spoken answers to text for NLP evaluation.
- Cloud storage service for storing recorded responses and generated PDF reports.
- Email/SMS gateway for notifications.

9.4 Communication Interfaces

- HTTPS/REST APIs between frontend, backend, and AI evaluation microservice.

10. System Features

10.1 AI-Driven Question Generation

Dynamically generates domain- and role-specific technical, HR, and coding questions with adjustable difficulty.

10.2 Automated Response Evaluation

Uses NLP/ML models to assess correctness, relevance, structure, and communication quality of candidate answers, producing a numeric score and qualitative feedback.

10.3 Performance Analytics Dashboard

Gives candidates a trend view of scores across attempts, and gives admins aggregate readiness analytics across candidates and domains.

10.4 Question Bank Management

Allows admins to curate and version the question bank and configure evaluation weightage per domain.

11. Assumptions and Constraints

11.1 Assumptions

- Candidates have access to a stable internet connection, a working webcam, and a microphone.
- The AI evaluation model has been pre-trained on a sufficiently representative dataset of interview Q&A.
- Institution/placement cell will provide an initial curated question bank to bootstrap the system.
- Users have basic digital literacy to navigate a web application.

11.2 Constraints

- Technical: AI evaluation accuracy is bounded by the quality/size of available training data and model latency limits.
- Operational: The system must run within the free/low-cost tiers of cloud and AI APIs available to a student project.
- Economic: Limited budget restricts use of premium proctoring or large-scale GPU inference for the current release.
- Legal: Candidate video/audio recordings and personal data must be handled per applicable data-protection norms.
- Time: The project must be delivered within the academic semester timeline defined for CSA10.

12. Validation of Requirement Completeness and Consistency

12.1 Stakeholder Coverage Check

Each stakeholder group identified in Section 4 maps to at least one functional requirement (candidates: FR-01–FR-07, FR-11–FR-12; admins: FR-08–FR-10), confirming that no stakeholder's needs are left unaddressed.

12.2 Ambiguity and Redundancy Check

- Requirement wording was reviewed to remove vague terms (e.g., 'fast', 'user-friendly') and replace them with measurable criteria (e.g., NFR performance target of 5 seconds).
- No two functional requirements were found to overlap in scope; each FR maps to a distinct use case.

12.3 Consistency Check

- Functional requirements (Section 5) are cross-checked against the use case table (Section 7.2) – every use case is backed by at least one FR, and every FR maps to a use case.

12.4 Identified Gaps and Suggested Improvements

- Add a requirement for candidate consent management before recording video/audio, to strengthen legal compliance.
- Add an explicit non-functional requirement for AI model bias/fairness auditing across candidate demographics.
- Define a formal SLA for AI evaluation turnaround time under peak load, beyond the current 5-second target.
- Add a requirement for offline/low-bandwidth fallback (text-only interview mode) to improve accessibility.

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